INORGANIC CUMULATIVE EXAMINATION

April 6, 1991

1. The high-spin d^4 complex $[Cr(H_2O)_6]^{2+}$ is *labile,* but the low-spin d^4 complex ion $[Cr(CN)_6]^{4-}$ is *inert.* Explain.

2. Why is the existence of a series of entering groups with different rate constants evidence for an associative mechanism (A or I_a)?

3. Is the reaction $[Co(NH_3)_6]^{3+} + [Cr(H_2O)_6]^{2+}$ likely to proceed by an inner-sphere or outer-sphere mechanism? Explain your answer.

4. The rate constants for the exchange reaction

$$CrX^{2+} + {}^*Cr^{2+} \rightarrow {}^*CrX^{2+} + Cr^{2+}$$

where *Cr is radioactive ^{51}Cr, are given in the table for reactions at 0°C and 1 *M* $HClO_4$. Explain the differences in the rate constants in terms of the probable mechanism of the reaction.

X^-	$k(M^{-1}\,s^{-1})$
F^-	1.2×10^{-3}
Cl^-	11
Br^-	60
NCS^-	1.2×10^{-4} (at 24°C)
N_3^-	>1.2

Department of Chemistry and Biochemistry
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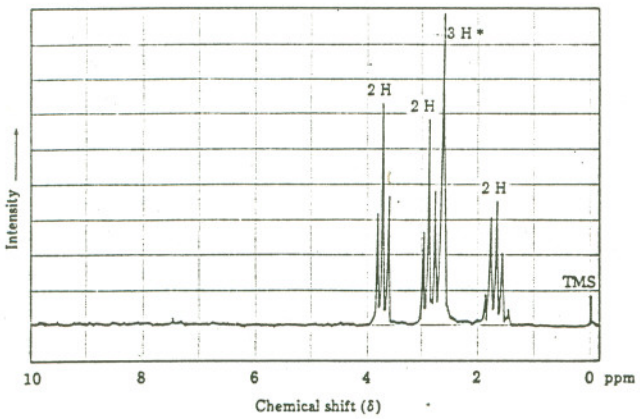
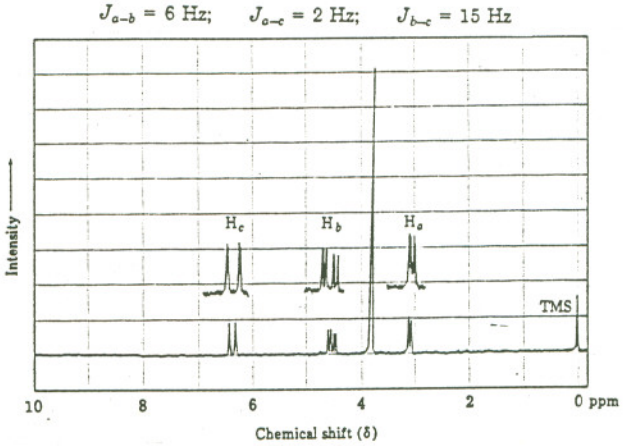
Organic Cumulative Examination, 4/6/1991

The ^{1}H-NMR spectra below are taken from the sophomore organic chemistry text by McMurry (problem references given). Identify the compounds whose spectra are given by writing their structural formulas. Do 10 of the 12--tell which two you want omitted. If you want a chance at partial credit for incorrect or incomplete answers, show your reasoning.

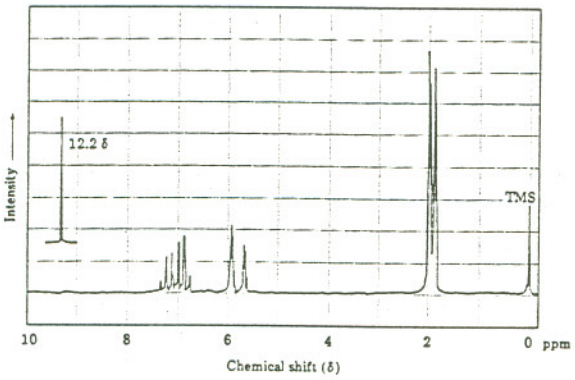
1 C_3H_9NO (25.50b)



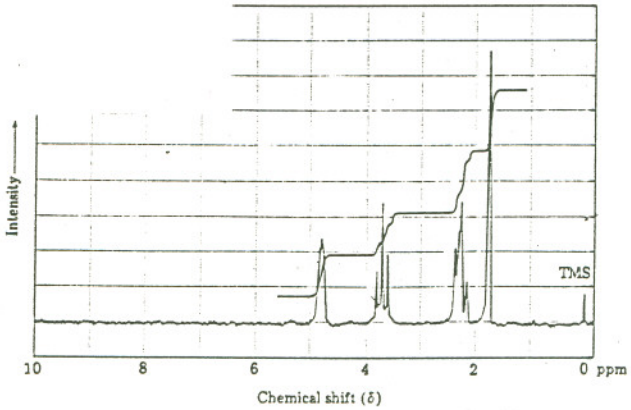
2 C_6H_6O (13.45)



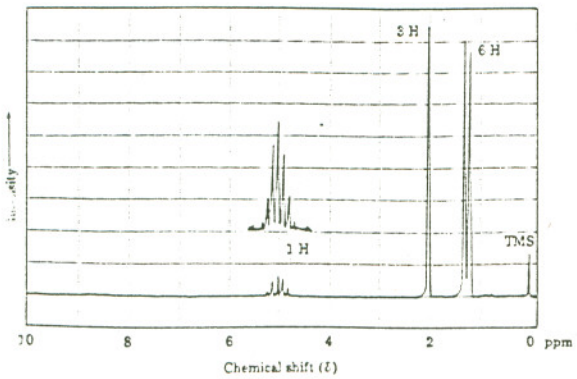
3 $C_4H_6O_2$ (20.39)



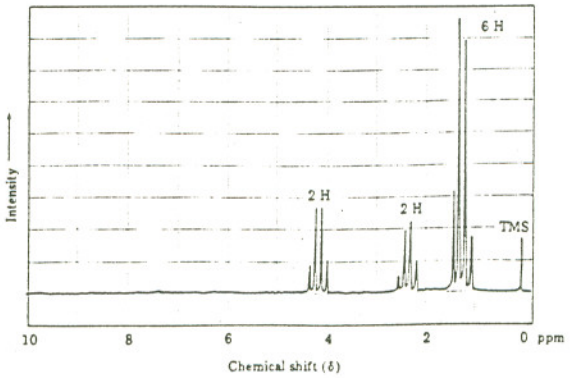
4 $C_5H_{10}O$ (17.41)



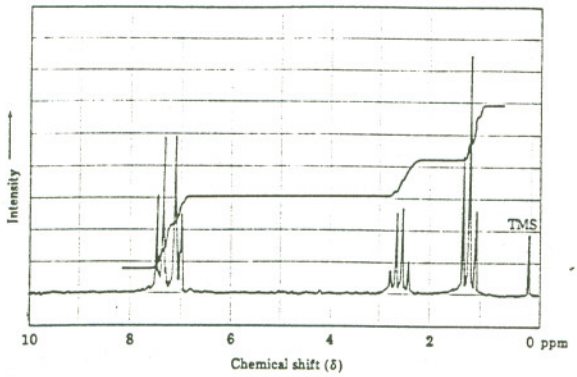
5 $C_5H_{10}O_2$ (21.62c)



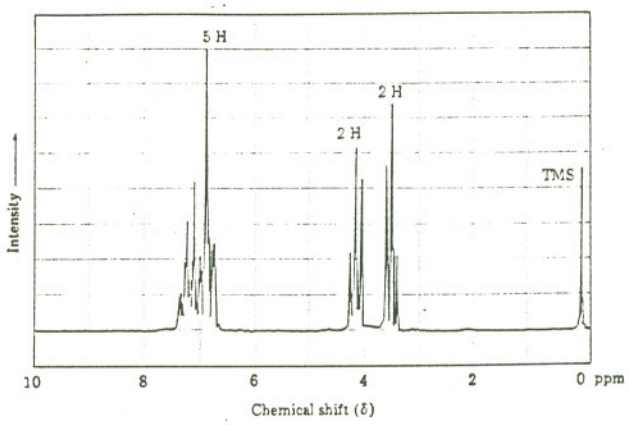
6 $C_5H_{10}O_2$ (21.61)



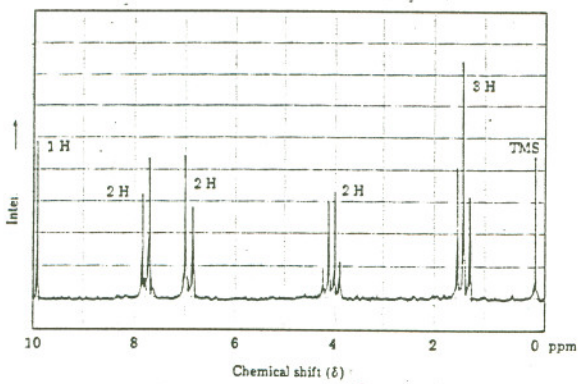
7 C_8H_9Br (13.47)



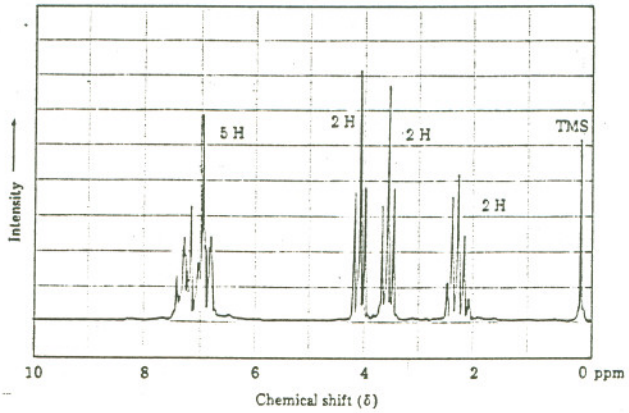
8 C_8H_9Br (13.48c)



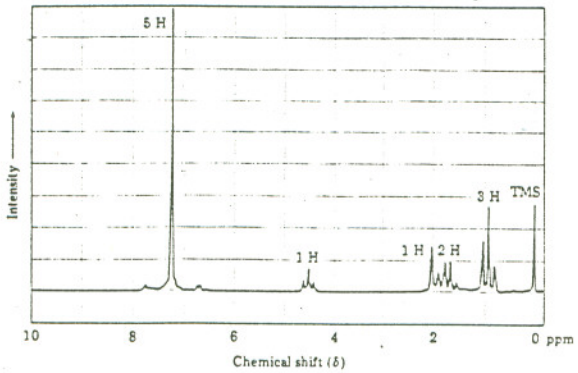
9 $C_9H_{10}O_2$ (19.56c)



10 $C_9H_{11}BrO$ (18.45a)



11 $C_9H_{12}O$ (17.47a)



12 $C_{10}H_{13}NO_2$ (26.39)

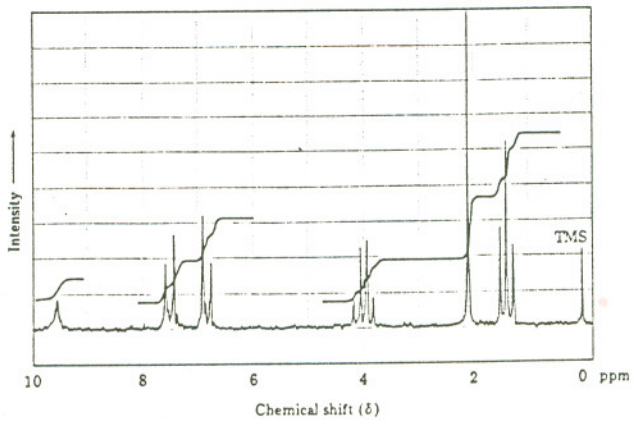


Table 13.4 Correlation of ^{1}H chemical shift with environment

Type of proton	*Formula*	*Chemical shift (δ)*
Reference peak	$(CH_3)_4Si$	0
Saturated primary	$-CH_3$	0.7–1.3
Saturated secondary	$-CH_2-$	1.2–1.4
Saturated tertiary	$>CH-$	1.4–1.7
Allylic primary	$>C=C-CH_3$	1.6–1.9
Methyl ketones	$-C(=O)-CH_3$	2.1–2.4
Aromatic methyl	$Ar-CH_3$	2.5–2.7
Alkyl chloride	$Cl-CH<$	3.0–4.0
Alkyl bromide	$Br-CH<$	2.5–4.0
Alkyl iodide	$I-CH<$	2.0–4.0
Alcohol, ether	$-O-CH<$	3.3–4.0
Alkynyl	$-C\equiv C-H$	2.5–2.7
Vinylic	$>C=C-H$	5.0–6.5
Aromatic	$Ar-H$	6.5–8.0
Aldehyde	$-C(=O)-H$	9.7–10.0
Carboxylic acid	$-C(=O)-O-H$	11.0–12.0
Alcohol	$-CH<O-H$	Extremely variable (2.5–5.0)

(*left* on the NMR chart). Everything we have learned about ^{13}C chemical shifts is also applicable to proton shifts. Proton chemical shifts tell a great deal about the chemical (magnetic) environments within a molecule.

Proton chemical shifts are expressed in δ units (recall 1 δ = 1 ppm of spectrometer frequency), just as for ^{13}C shifts. In contrast to ^{13}C shifts, however, proton chemical shifts fall within the rather narrow range of 0–10 δ. The exact spot within this range is highly characteristic of environment, as shown in Table 13.4 and Figure 13.16.

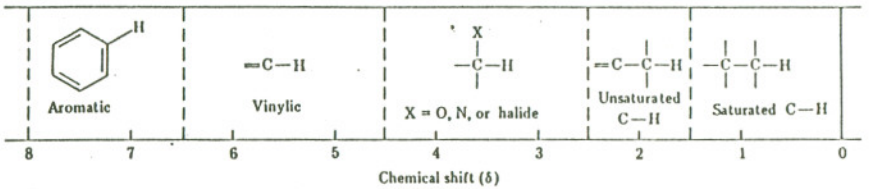


Figure 13.16 Chemical shifts for different kinds of protons

The majority of ^{1}H NMR absorptions occur from 0 to 8 δ, a range that can be conveniently divided into five regions, as shown in Table 13.5. By memorizing the positions of these five regions, it's often possible to tell at a glance what general kinds of protons a molecule contains.

Table 13.5 Regions of the ^{1}H NMR spectrum

Region (δ)	*Proton type*	*Comments*
0–1.5	$-CH<-CH<$	Protons on carbon next to saturated centers absorb in this region. Thus the alkane portions of most organic molecules show complex absorption here.
1.5–2.5	$=CH-CH<$	Protons on carbon next to unsaturated centers (allylic, benzylic, next to carbonyl) show characteristic absorptions in this region, just downfield from other alkane resonance.
2.5–4.5	$X-CH<$	Protons on carbon next to electronegative atoms (halogen, O, N) are deshielded because of the electron-withdrawing ability of these atoms. Thus the protons absorb in this midfield region.
4.5–6.5	$>C=C<$ H	Protons on double-bond carbons (vinylic protons) are strongly deshielded by the neighboring pi bond and therefore absorb in this characteristic downfield region.
6.5–8.0	H	Protons on aromatic rings (aryl protons) are strongly deshielded by the pi orbitals of the ring and absorb in this characteristic low-field range.